

Unit 8, Lesson 1: Area of Triangles – Part 1

Benchmark

MA.6.GR.2.1 Derive a formula for the area of a right triangle using a rectangle. Apply a formula to find the area of a triangle.

Learning Target

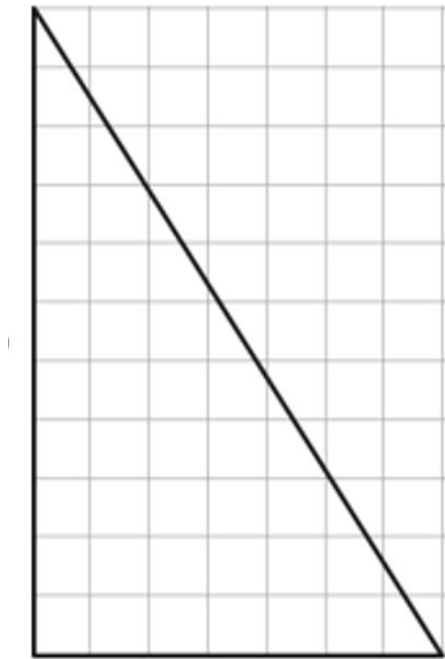
- ☐ I can discover the formula for the area of a triangle.

8.1.1 Warm-Up: The Importance of Struggle

When learning something new, whether it is a sport, a musical instrument, or even math, you may experience failure before you experience success. The key is to keep going even when you make mistakes. Learning takes place during the time you practice and during the time you struggle.

1. Which choice would you feel less comfortable doing?
 - Answering a question and getting it wrong.
 - Saying, “I do not know,” as your answer to a question.
2. Describe a time that you persevered when learning something new.
3. When you persevered in learning, how did you overcome any initial fear(s)?

8.1.2 Exploration: Discovering the Area of a Right Triangle



A **right triangle** is shown. Each square of the grid paper is 1 centimeter (cm) by 1 cm.

- 1. Trace two copies of the right triangle on paper, then cut them out.
- 2. Form a rectangle using the two right triangles. Paste the triangles here.

3. Using the image just created, complete the table.

	Rectangle	Triangle
Height	_____ cm	_____ cm
Base	_____ cm	_____ cm

4. What is the area of the rectangle? Include units.

5. Describe the relationship between the area of one right triangle and the area of the rectangle.

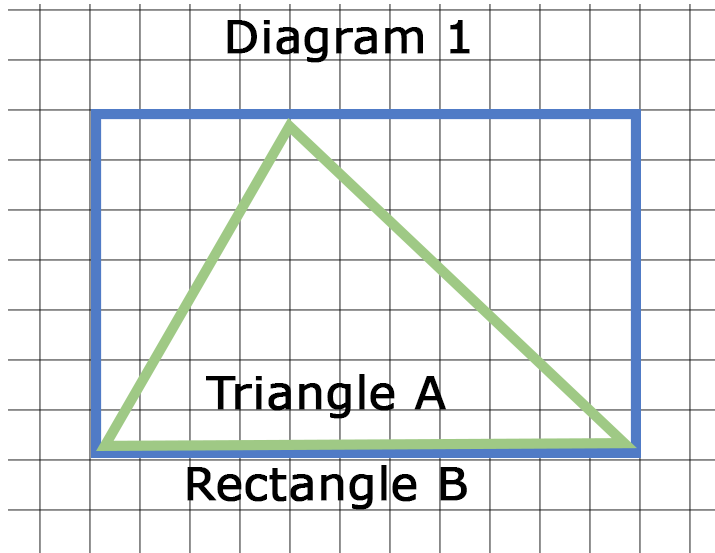
6. What is the formula for the area of a rectangle?

7. Use the relationship observed and the formula for the area of a rectangle to write a formula that can be used to find the area of a right triangle.

8.1.3 Collaboration: Area of an Acute and an Obtuse Triangle

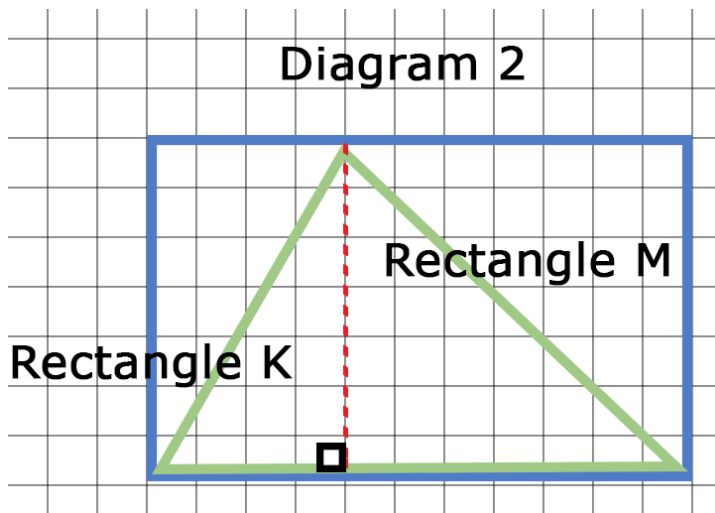
An **acute triangle** is shown. Each square of the grid paper in Diagram 1 and Diagram 2 is 1 centimeter (cm) by 1 cm.

Diagram 1



Rectangle B is drawn around acute Triangle A, as shown.

Diagram 2



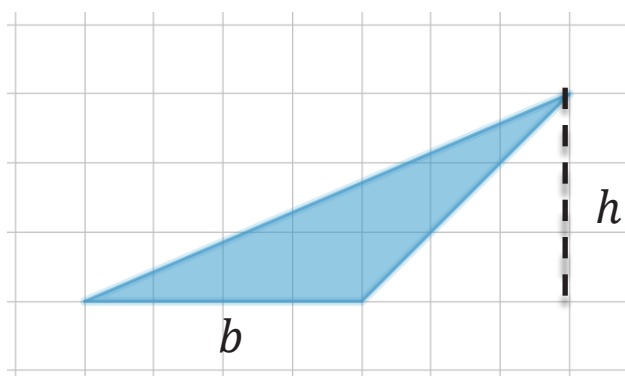
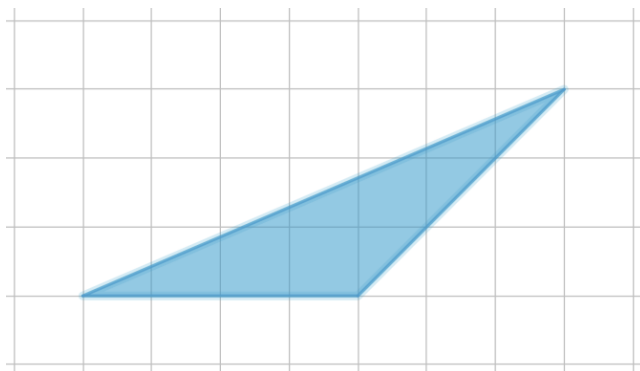
The height of acute Triangle A is shown as a dashed red line. It divides Rectangle B into two smaller rectangles, Rectangle K and Rectangle M, and divides Triangle A into two smaller right triangles.

1. Use the two diagrams to complete the table.

	Area of the Rectangle	Area of the Right Triangle Drawn Inside the Rectangle
Rectangle K		
Rectangle M		

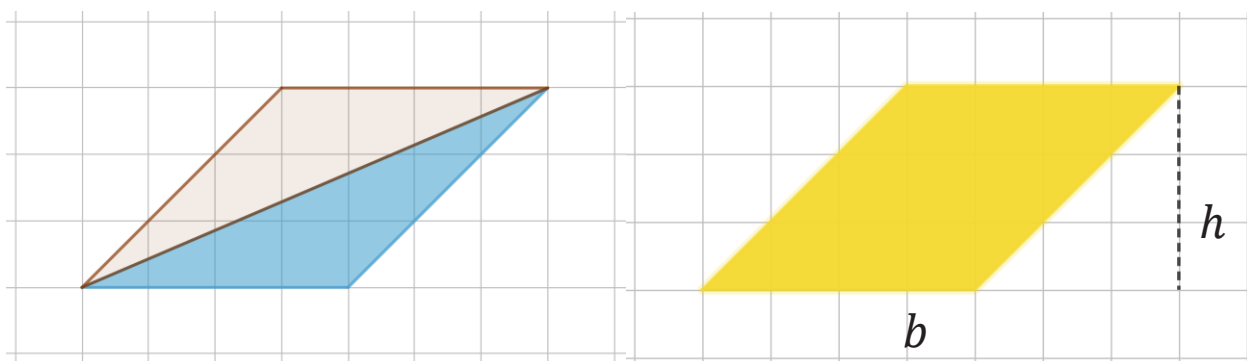
2. The area of acute Triangle A is equal to the sum of the right triangle inscribed in Rectangle K and the right triangle inscribed in Rectangle M.
- a. What is the area of acute Triangle A?
 - b. What is the area of Rectangle B?
 - c. What is the relationship between the area of Rectangle B and the area of acute Triangle A?

3. An **obtuse triangle** is shown. Each square of the grid paper is 1 centimeter (cm) by 1 cm.

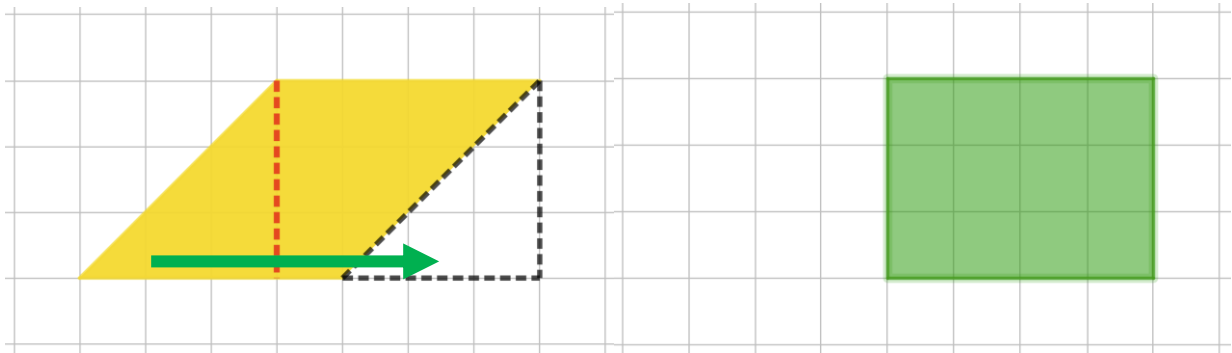


- a. What is the length of the triangle's base (b)?
- b. What is the height (h) of the triangle?

4. A copy of the triangle is made, and the two triangles are arranged to form a parallelogram with a base of 4 cm and a height of 3 cm, as shown.



The parallelogram is then decomposed, and the parts are rearranged to form a rectangle.



a. Complete the table.

	Rectangle	Obtuse Triangle
Height	_____ cm	_____ cm
Base	_____ cm	_____ cm

- b. How many obtuse triangles were needed to create the rectangle?
- c. What is the area of the rectangle?
- d. What is the relationship between the area of the rectangle and the obtuse triangle?
- e. Based on this relationship, what is the area of the obtuse triangle?

8.1.4 Bring It Together: Formula for the Area of a Triangle

1. Complete the statements.

The area of the **right** triangle was

- ☐ half
- ☐ the same as

 the area of the rectangle with the same base and height.

The area of the **acute** triangle was

- ☐ half
- ☐ the same as

 the area of the rectangle with the same base and height.

The area of the **obtuse** triangle was

- ☐ half
- ☐ the same as

 the area of the rectangle with the same base and height.

2. The formula for the area of any rectangle is $A = bh$.

$$A = \text{base} \cdot \text{height}$$

The formula for the area of any triangle is $A = \underline{\hspace{1cm}}bh$.

$$A = \underline{\hspace{1cm}}(\text{base} \cdot \text{height})$$

8.1.5 Wrap-Up: Reflection

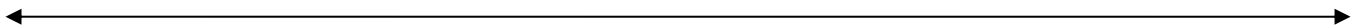
1. Place an **X** on the line to indicate your mastery of today's lesson.

I can explain why the area of a triangle is half the area of a rectangle with the same base and height for right triangles, acute triangles, and obtuse triangles.

I need help.
I can't get started.

I'm getting
there.

I understand and
have evidence.



2. Write a brief explanation of your placement of the X.